

Generative AI in Politics Research

Lecture Week 11

Dr Zach Dickson & Dr Dan De Kadt

London School of Economics

WT 2025

Overview

Data Science
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Generative AI as a
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Simulating Human
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Generative AI

What do we mean by Generative AI?

ChatGPT



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Generative AI

- ▶ Generative AI refers to a class of algorithms that can generate new content, such as text, images, or audio, based on patterns learned from existing data.
- ▶ Generative AI is a subset of artificial intelligence (AI) that focuses on creating new content rather than just analyzing or processing existing data.
- ▶ Generative AI models are trained on large datasets and use machine learning techniques to understand the underlying patterns and structures of the data.
- ▶ Generative AI can be used in various applications, including natural language processing (NLP), computer vision, and audio synthesis.

Generative AI in Research

- ▶ Research assistant
- ▶ As a tool for methods/analysis
- ▶ Data collection/generation
- ▶ Simulation of human behaviour
- ▶ Subject of study

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Limitations of
Generative AI

- ▶ Less formalized/discussed
- ▶ Generative AI can be used to assist researchers in various tasks, including:
 - ▶ Feedback on writing
 - ▶ Coding
 - ▶ Data analysis
 - ▶ Writing and editing
 - ▶ Idea generation
- ▶ Real-world relevance

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Generative AI as an research tool

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Limitations of
Generative AI

- ▶ Many of the most popular generative AI tools are based on large language models (LLMs)
- ▶ LLMs can be used for a variety of tasks, including:
 - ▶ translation
 - ▶ summarization
 - ▶ classification

Zero-shot classification

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PNAS

BRIEF REPORT

POLITICAL SCIENCES

 OPEN ACCESS

ChatGPT outperforms crowd workers for text-annotation tasks

Fabrizio Gilardi ¹, Meysam Alizadeh , and Maël Kubli 

Edited by Mary Waters, Harvard University, Cambridge, MA; received March 27, 2023; accepted June 2, 2023

July 18, 2023 | 120 (30) e2305016120 | <https://doi.org/10.1073/pnas.2305016120>

Many NLP applications require manual text annotations for a variety of tasks, notably to train classifiers or evaluate the performance of unsupervised models. Depending on the size and degree of complexity, the tasks may be conducted by crowd workers on platforms such as MTurk as well as trained annotators, such as research assistants. Using four samples of tweets and news articles ($n = 6,183$), we show that ChatGPT outperforms crowd workers for several annotation tasks, including relevance, stance, topics, and frame detection. Across the four datasets, the zero-shot accuracy of ChatGPT exceeds that of crowd workers by about 25 percentage points on average, while ChatGPT's intercoder agreement exceeds that of both crowd workers and trained annotators for all tasks. Moreover, the per-annotation cost of ChatGPT is less than \$0.003—about thirty times cheaper than MTurk. These results demonstrate the potential of large language models to drastically increase the efficiency of text classification.

ChatGPT | text classification | large language models | human annotations | text as data

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Zero-shot classification

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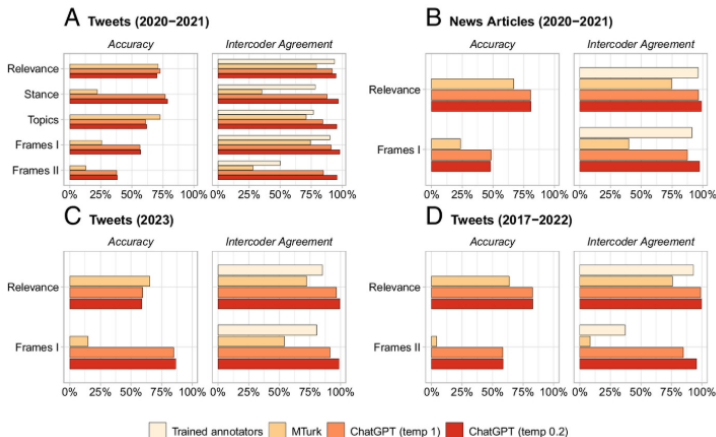
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Translation/Summarization

The screenshot shows the UKIP website header with the logo and navigation links. The main article title is 'Energy Security & Net Zero' by Antony Nailer, dated 13 February 2023. The article text discusses Grant Shapps' role in government and lists five key points regarding energy security and net zero. A sidebar on the right lists recent national news items.

UKIP
THE NEW RIGHT

JOIN HERE | SITEMAP

Energy Security & Net Zero

Antony Nailer • 13 February 2023

News & Social Media / Post

Search the news

Antony Nailer
Energy Spokesman for the UK Independence Party

Grant Shapps has been in government quite a long time and as well as being Deputy Chairman of the Conservative Party he has also had the following appointments.

1. Minister for state for International Development, May 2015 till November 2015. This department finds ways to throw taxpayers money at foreign governments.
2. Secretary of State for Transport, July 2019 till September 2022. He used the Covid period to have towns in his constituency and elsewhere block off through roads with planters making them into cul-de-sacs, to continue the war against the motorist.
3. Secretary of State for the Home office under Liz Truss 19-25 October 2022.
4. Secretary of State for Business, Energy, and Industrial Strategy, (BEIS).
5. This had an emphasis on building more wind farms which continued the decline in granting licences and planning permissions for mining of coal, drilling for oil, and included a moratorium for fracking for gas. Now we have to import these things at enormous expense, if they are available, and with the Russian gas turned off we will need to import enormous amounts of Liquefied Natural Gas from the USA to prevent gas rationing.

Rishi Sunak has done a mini reshuffle, a bit like tidying the deckchairs on Titanic after it hit the iceberg, and he has now put the fox into the chicken run. Grant Shapps has become Secretary of State for Energy Security and Net Zero. This is a classic oxymoron, where words have conflicting meaning, such as 'Army Intelligence'.

Recent National News

- Enough is Enough
- Ever wondered what makes UKIP different?
- The Father of BREXIT Dies
- They are certainly 'Changing Britain'
- The Power of Labour Suggestion
- Reform isn't Working (even with its closest
- General Election 2024 - Our Mission
- UKIP & Reform UK - What's the Difference?
- The bill for PP is Unustainable
- Stammer's Energy Security Policy

View More

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Dickson & Hobolt (2024)

Model output: *The primary issue addressed in the text is the government's approach to reducing pollution from car use and achieving Net Zero. The party's position on this issue is against. The text criticizes the government for prioritizing more renewables and less fossil fuels in their efforts to achieve this goal, arguing that it will increase energy insecurity and reliability, drive up costs, and waste taxpayers' money.*

Coding Open-text Responses

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Mellon et al (2024)

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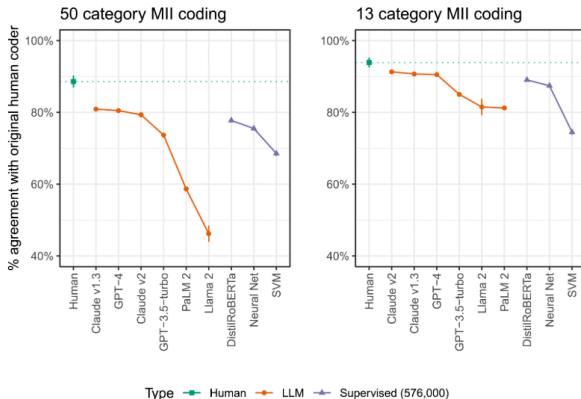


Figure 2. Percentage of responses that agree with original human coder across the *new test set* (text responses that did not appear in waves 1–20). Supervised models trained on 576,000 BESIP waves 1–20 cases.

Classifying sentiment of political tweets

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Ornstein, Blasingame, & Truscott (*unpublished*)

Table 3. Sample of tweets with ambiguous language

Tweet	Authors	GPT-3	Dictionary	Naïve Bayes	TweetNLP
I'm sure the news media will provide sober, nuanced analysis of Masterpiece Cake Shop. #SCOTUS	Negative	Negative	Positive	Positive	Neutral
Gonna have a drink at "the low bar" to celebrate SCOTUS ruling that laws sometimes apply to the president.	Negative	Positive	Positive	Positive	Positive
SCOTUS kicked the can down to [sic] road!	Negative	Negative	Neutral	Negative	Neutral
What a nice Supreme Court decision to lead off pride month. So excited to see all this legal discrimination about to happen. Want to thank two of my heroines RBG and Sotomayor for dissenting.	Negative	Negative	Positive	Positive	Positive

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Two applications of Generative AI in survey research:

- ▶ Generative AI in survey experiments
- ▶ Generative AI as an interviewing tool

Adaptive surveys

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Uses LLMs to generate survey questions that adapt to user input

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FIGURE 1. Dynamic Tailored Conjoint Design

(a) Open-ended question

Thinking about issues that define the American political system, what is an issue that you care deeply about and what is your position on that issue?

Please write a sentence or two about an issue that you care about and where you stand on the issue.

For example, if you care about farm subsidies, you can write "I believe farm subsidies should be increased to help farmers."

I believe people need to learn more re CRT, not less. Hiding from uncomfortable subjects doesn't make us happier, it makes us realize how weak our collective backbone is. Who could be happy living like that?



(b) Personalized Conjoint

	Candidate 1	Candidate 2
Partisanship	Democrat	Democrat
Race/Ethnicity	Asian American	Black
Asylum Policy	Allow political refugees to obtain asylum.	Allow economic refugees to obtain asylum.
Religion	Catholic	Presbyterian
Career	State representative	Agriculture
Veteran Status	Served	Did Not Serve
Critical Race Theory	The amount of education on Critical Race Theory should remain the same.	People should learn more about Critical Race Theory, not less.
Deportation Policy	Identify and deport all undocumented immigrants.	Identify and deport all undocumented immigrants.
Social Spending	Decrease spending on social programs that assist immigrants.	Keep spending on social programs that assist immigrants the same.
Sex	Female	Male
Border Wall Policy	Build a physical wall along the entire southern border.	Build a physical wall along parts of the southern border.
Age	62	66

Note: Screenshots of the survey design. (a) Participants begin by responding to an open-ended question about their most important issue, which is used to generate a personalized conjoint task. (b) Participants complete a set of candidate choice trials with their tailored issue position featured alongside immigration issue positions and demographic characteristics. See Appendix A for more details on conjoint task.

Velez, Yamil. "Do Personal Issue Priorities Trump Group Policies? Exploring the Impact of Deeply-Held Issues among Latinos using Personalized Conjoint Experiments"

Language Models in Interviewing

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Conversations at Scale: Robust AI-led Interviews with a Simple Open-Source Platform*

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November 20, 2024

Abstract

The advent of large language models (LLMs) provides an opportunity to conduct qualitative interviews at a large scale, with thousands of respondents, creating a bridge between qualitative and quantitative methods. In this paper, we develop a simple, versatile open-source platform for researchers to run AI-led qualitative interviews. Our approach incorporates established best practices from the sociology literature, uses only a single LLM agent with low latency, and can be adapted to new interview topics almost instantaneously. We assess its robustness by drawing comparisons to human experts and using several respondents-based quality metrics. Its versatility is illustrated through four broad classes of applications: eliciting key factors in decision making, political views, views of the external world, and subjective mental states. High performance ratings are obtained in all of these domains. The platform is easy to use and deploy: we provide detailed explanations and code for researchers to swiftly set up and test their own AI-led interviews. In addition, we develop, validate, and share a simple LLM-based pipeline for textual analysis and coding of large volumes of interview transcripts.

Keywords: qualitative interviews; large language models; surveys.

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Geicke & Jaravel (2024)

Table XI Quality Metrics for the AI-led Interview on Electoral Choices

	Fraction of Respondents
<i>In the future, would you rather take the interview with</i>	
... An AI	35.90%
... A human	15.38%
... I do not mind	48.72%
<i>How well does it summarize your reasons?</i>	
1 ("poorly") to 4 ("very well")	3.78 (0.41)

Notes: This table reports two measures of perceived quality for the AI-led interview on educational and occupational choices.

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Limitations of
Generative AI

- ▶ Generative AI can be used to simulate human behaviour in various contexts, including:
 - ▶ Social media interactions
 - ▶ Survey responses
 - ▶ Game theory

Simulating Human Behaviour

Examples:

- ▶ Simulated human samples (Argyle et al 2023)
 - ▶ But, debate over reliability of synthetic responses / "silicon samples" (See Bisbee et al 2023)
- ▶ Behavioural game theory applications (Li et al 2023; Akata et al 2023)
- ▶ Simulated social media environment (using LLMs to interact w/ respondents) (Allamong et al 2024)
- ▶ LLM performance in wargames (Lamparth et al 2024)
- ▶ Do humans or LLMs make more convincing political arguments? (Palmer & Spirling 2024)

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- ▶ Generative AI – LLMs – are the focus of a growing body of research
- ▶ Research focuses on the implications of generative AI in:
 - ▶ Misinformation
 - ▶ Persuasion
 - ▶ Microtargeting
 - ▶ Campaigning

Microtargeting and Persuasion

Hackenburg & Margetts (2024)

PNAS

RESEARCH ARTICLE

POLITICAL SCIENCES

 OPEN ACCESS

Evaluating the persuasive influence of political microtargeting with large language models

Kobi Hackenburg ¹ and Helen Margetts 

Edited by Matthew Jackson, Stanford University, Stanford, CA; received February 13, 2024; accepted May 8, 2024

June 7, 2024 | 121 (24) e2403116121 | <https://doi.org/10.1073/pnas.2403116121>

Significance

Advances in large language models (LLMs) have raised concerns over scalable, personalized political persuasion. Here, we integrate user data into GPT-4 prompts in real-time, facilitating the live creation of messages tailored to persuade individual users on political issues. We then deploy this application at scale to test whether personalized, microtargeted messaging offers a persuasive advantage compared to nontargeted messaging. We find that while messages generated by GPT-4 were persuasive, in aggregate, the persuasive impact of microtargeted messages was not statistically different from that of nontargeted messages. These findings suggest—contrary to widespread speculation—that the influence of current LLMs may reside not in their ability to tailor messages to individuals but rather in the persuasiveness of their generic, nontargeted messages.

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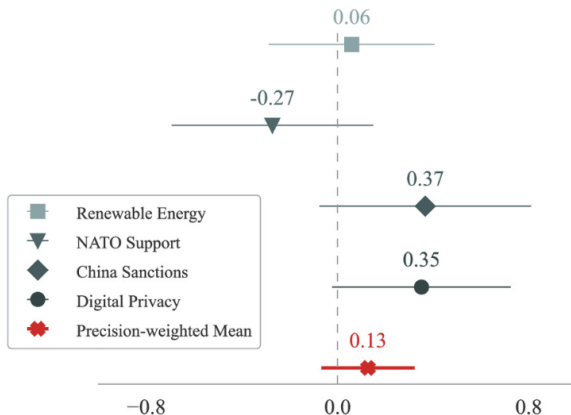
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Microtargeting and Persuasion

Hackenburg & Margetts (2024)



**Estimated Change in Persuasive Influence
per Additional Attribute Targeted
in Percentage Points (95% CI)**

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- ▶ Generative AI can produce biased or inaccurate content if the training data is not representative or contains biases
- ▶ Generative AI models can be difficult to interpret, making it challenging to understand how they arrive at specific outputs
- ▶ Generative AI models can be vulnerable to adversarial attacks, where malicious actors manipulate the input data to produce misleading or harmful outputs
- ▶ Generative AI models can raise ethical concerns, such as the potential for misuse in creating deepfakes or spreading misinformation

Is ChatGPT Politically Biased?

Motoki, Pinho Neto & Rodrigues (2024)

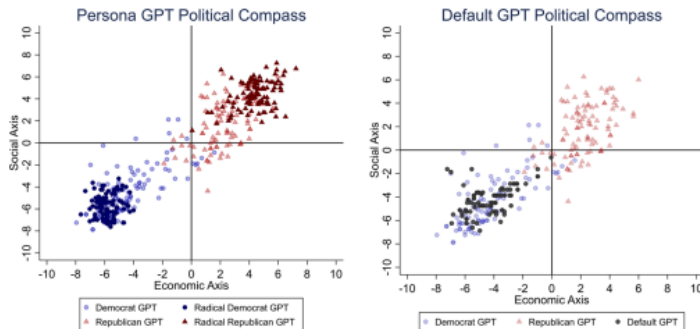


Fig.2 Political Compass quadrant—Average and Radical ChatGPT Impersonations (left) and Default and Average ChatGPT Impersonations (right). *Notes:* Political Compass quadrant classifications of the 100 sets of answers of each impersonation. The vertical axis is the social dimension: more negative values mean more libertarian views, whereas more positive values mean more authoritarian views. On the horizontal axis is the economic dimension: more negative values represent more extreme left views, and more positive values represent more extreme right views

Can LLMs Represent Public Opinion?

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Bisbee et al (2024)

"It is [YEAR]. You are a [AGE] year-old, [MARST], [RACETH] [GENDER] with [EDUCATION] making [INCOME] per year, living in the United States. You are [IDEO], [REGIS] [PID] who [INTEREST] pays attention to what's going on in government and politics."¹³

In each query to ChatGPT, the characteristics in brackets are substituted with values corresponding to a real respondent in the 2016 or 2020 wave of the ANES. These values include:

- [YEAR]: 2016 or 2020
- [AGE]: age in years of ANES respondent
- [RACETH]: non-Hispanic white, non-Hispanic black, or Hispanic
- [GENDER]: male or female
- [MARST]: divorced, married, separated, single, or widowed
- [EDUCATION]: a high school diploma, some college but no degree, a bachelor's degree or more
- [INCOME]: \$30k, \$50k, \$80k, \$100k, \$150k or more
- [IDEO]: an extremely liberal, a liberal, a slightly liberal, a moderate, a slightly conservative, a conservative, an extremely conservative
- [REGIS]: registered, unregistered
- [PID]: Democrat, Independent, Republican
- [INTEREST]: never, sometimes, frequently, regularly, always

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Bisbee et al (2024)

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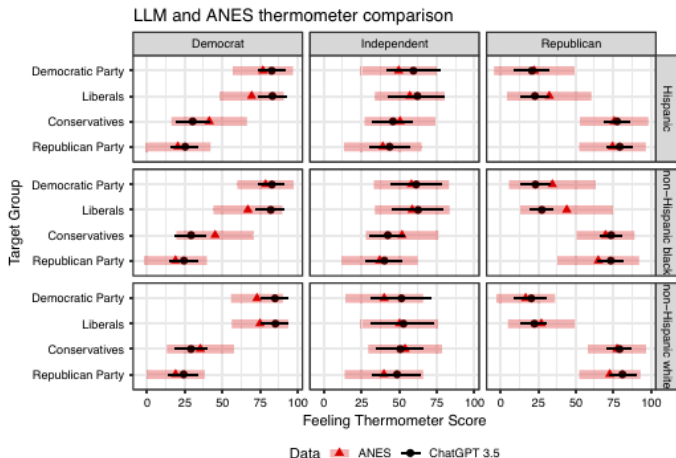


Figure 2. Average feeling thermometer results (x-axis) for different target groups (y-axes) by party ID of respondent (columns). Average ANES estimates from the 2016 and 2020 waves indicated with red triangles and one standard deviation indicated with thick red bars. LLM-derived averages indicated by black circles and thin black bars. Sample sizes for each groupwise comparison are identical.

Conclusion

- ▶ Generative AI is a powerful tool that can be used in various research contexts - your imagination is the limit
- ▶ LLMs are not perfect and can produce biased or inaccurate content
- ▶ Generative AI is a rapidly evolving field, and new applications and techniques are constantly being developed